

# Homework Report: 3D Geometric Reconstruction of a Cylindrical Vault

Image Analysis and Computer Vision (2025-2026)

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# Homework Requirements Summary

This report addresses all six homework requirements as follows:

1. **Vanishing Line of Plane Perpendicular to Cylinder Axis**
  - Location: Section 2, Figure 2
  - Result:  $\mathbf{l}'_{\infty}$  computed via cross product  $\mathbf{v}_v \times \mathbf{v}_t$
2. **Vanishing Point of Cylinder Axis Direction**
  - Location: Section 2, Experimental Results (Section 6)
  - Result:  $v_a = (8583.8, 3457.2)$  pixels
3. **Euclidean Rectification  $\mathbf{H}_R$  and Non-Apical Nodal Point**
  - Location: Section 3, Figure 3
  - Result:  $\mathbf{H}_R = \mathbf{H}_S \mathbf{H}_{affine}$  (stratified approach)
  - Non-apical point  $N_{12}$  shown in rectified image (yellow/red markers)
4. **Calibration Matrix  $\mathbf{K}$** 
  - Location: Section 4, Experimental Results (Section 6)
  - Result:  $f_x = 1871.4$  pixels,  $f_y = 2422.3$  pixels,  $(c_x, c_y) = (1693.3, 49.0)$  pixels
5. **3D Coordinates of Dozen Points from Diagonal Arc**
  - Location: Section 5, Table 1
  - Result: 12 points from Arc A1 listed in camera reference frame
6. **Cylinder Axis Localization w.r.t. Camera**
  - Location: Section 5, Figure 4
  - Result: Direction  $[0.9054, 0.3460, 0.2459]^T$ , Point  $[0.0690, 0.7618, 0.7242]^T$

## 1. Problem Formulation

**Objective:** Perform a metric 3D reconstruction of the “San Maurizio” church interior from a single uncalibrated image. The scene contains a cylindrical vault with straight lines and curved diagonal arcs (ogival ribs).

**Key Constraints:**

- **Geometric Primitives:** Two families of diagonal arcs ( $A$  and  $B$ ) and straight segments (vertical columns, horizontal ties).
- **Symmetry:** Arcs  $a_i$  and  $b_i$  are symmetric with respect to a vertical plane  $\pi_v$  parallel to the cylinder axis.
- **Metric Info:** The distance between neighboring arcs of the same family is  $d = 1$ .

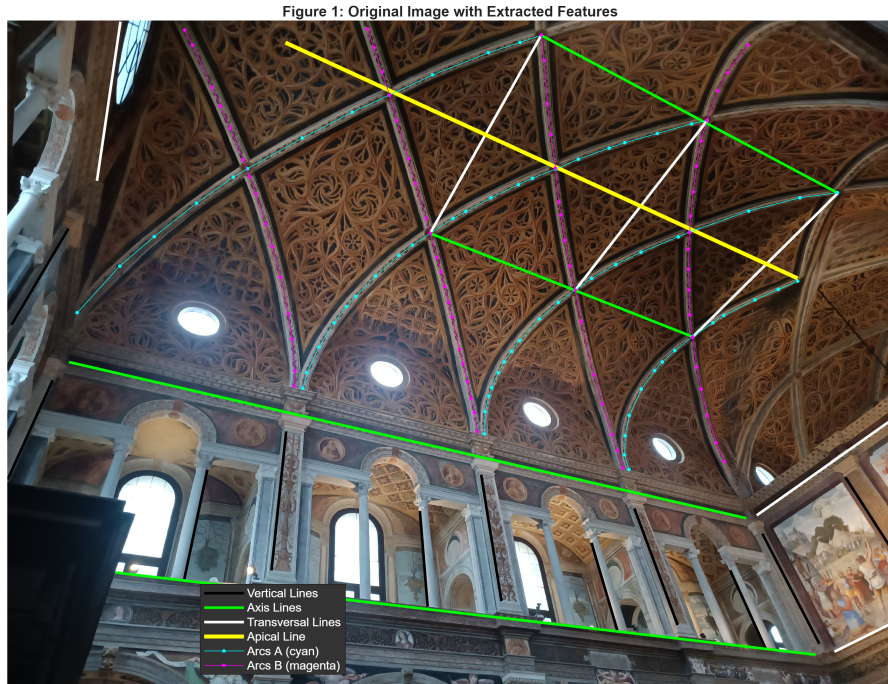


Figure 1: Original image of the San Maurizio church vault with extracted features: vertical lines (black), axis lines (green), transversal lines (white), apical line (yellow), and diagonal arcs of family  $A$  (cyan) and family  $B$  (magenta).

## 2. Vanishing Points and Vanishing Line

### 2.1. General Theory

In single-view geometry, parallel lines in 3D space intersect at a “Vanishing Point” (VP) in the 2D image plane [2].

- **Orthogonality:** If three directions in space are mutually orthogonal (e.g., X, Y, Z axes), their vanishing points ( $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ ) provide strong constraints on the camera geometry.
- **Vanishing Line ( $\mathbf{l}'_\infty$ ):** The vanishing line of any plane perpendicular to the cylinder axis is denoted as  $\mathbf{l}'_\infty$ . It represents the projection of the line at infinity for that specific set of parallel planes.

## 2.2. Application to San Maurizio & Implementation

**Application:** We identified three principal orthogonal directions in the church:

1. **Vertical ( $v_v$ ):** Defined by columns and walls.
2. **Axis ( $v_a$ ):** Defined by the horizontal ties connecting the arches (parallel to the cylinder axis).
3. **Transversal ( $v_t$ ):** Defined by the lines connecting opposite columns (perpendicular to the axis).

**Code Flow:**

1. **Feature Loading:** The script loads `lines_v`, `lines_axis`, and `lines_trans` from `features_fixed.mat`.
2. **SVD Estimation:** We calculate each VP using Singular Value Decomposition (SVD) on the line segment endpoints to minimize algebraic error.
  - *Code Reference:* `[~, ~, V] = svd(A_vert);` (similarly for axis/transversal).
3. **Vanishing Line:** The code computes the cross product  $\mathbf{l}'_\infty = \mathbf{v}_v \times \mathbf{v}_t$  to find the vanishing line of the vertical planes  $\pi_{ij}$  perpendicular to the cylinder axis.
  - *Code Reference:* `vanishing_line = cross(v_vert_pixel, v_trans_pixel);`

**Critical Observation - Vanishing Line Position:**

As visible in Figure 2, the vanishing line  $\mathbf{l}'_\infty$  cuts through the upper portion of the image. This has important consequences:

- **Points Near the Line:** Image points close to  $\mathbf{l}'_\infty$  will be mapped to extremely large coordinates (approaching infinity) during affine rectification, since the transformation  $\mathbf{H}_{affine}$  sends this line to infinity.
- **Numerical Stability:** The homogeneous coordinate divisor  $H(3, :) \cdot p$  approaches zero for points on or near  $\mathbf{l}'_\infty$ , causing numerical instability and extreme stretching.
- **Image Content Loss:** Features lying above the vanishing line (e.g., upper portions of the vault) cannot be reliably rectified due to this singularity.

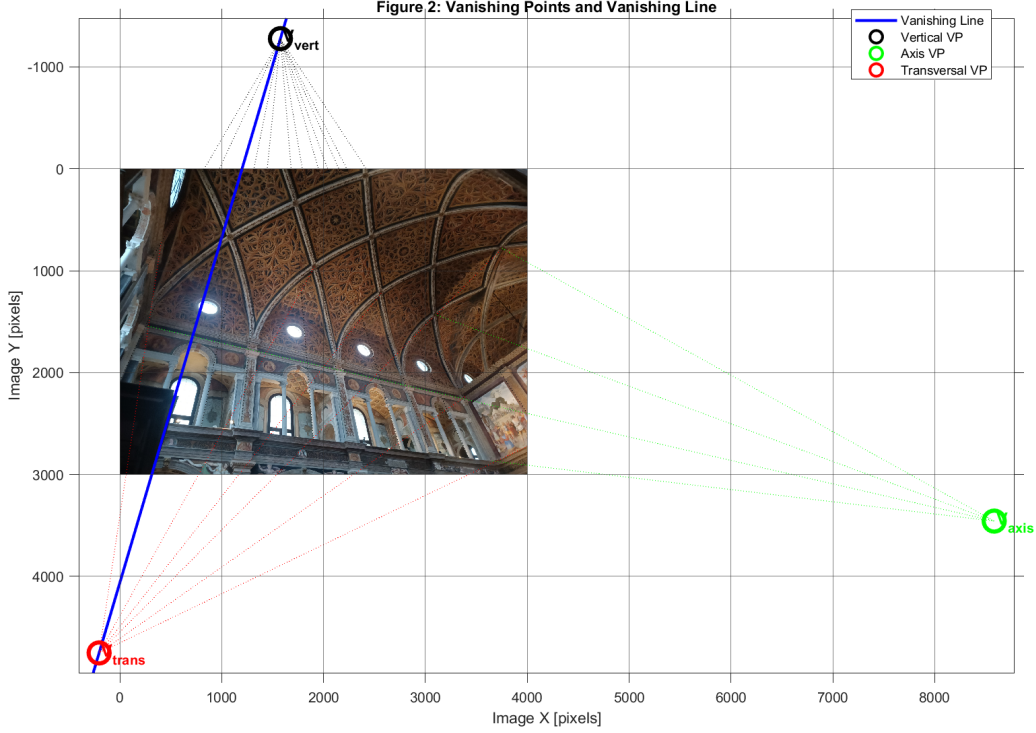


Figure 2: Vanishing points and vanishing line visualization. The three orthogonal vanishing points (vertical in black, axis in green, transversal in red) are shown with their corresponding convergence lines. The blue line represents the vanishing line of the vertical plane.

### 3. Euclidean Rectification

#### 3.1. General Theory: Stratified Reconstruction

Stratified reconstruction is a general method for recovering Euclidean geometry from a single perspective image by progressively upgrading the geometric structure [1, 3]. The approach identifies and recovers the projective entities that encode geometric invariants.

##### Projective Geometry Foundation:

In the projective plane  $\mathbb{P}^2$ , Euclidean geometry is recovered by identifying two special entities:

1. The **Line at Infinity** ( $l_\infty$ ): This line contains all points at infinity (directions). Affine transformations preserve this line, maintaining properties like parallelism.
2. The **Circular Points at Infinity** ( $I, J$ ): These are the two complex conjugate points on  $l_\infty$  where all circles intersect. Euclidean transformations (similarities) preserve these points, maintaining angles and shape. The Dual Conic to the Circular Points ( $C_\infty^*$ ) is defined as  $C_\infty^* = IJ^T + JI^T$ .

### The Stratification Hierarchy:

1. **Affine Rectification:** Identify the vanishing line  $l'_\infty$  (the image of  $l_\infty$ ) and map it to its canonical position  $[0, 0, 1]^T$  [1]. This restores affine invariants: parallelism, ratios of areas, and ratios of lengths along parallel lines.
2. **Metric Rectification:** Identify the image of the Dual Conic to the Circular Points ( $C_\infty^*$ ). This can be achieved through:
  - **Orthogonality constraints:** Enforcing that known orthogonal directions become perpendicular
  - **Circular/spherical constraints:** Enforcing that known circles or spheres project correctly
  - **Equal length constraints:** Enforcing known metric relationships

According to Lecture C, a transformation is a similarity (Euclidean) if and only if it leaves the circular points  $I, J$  invariant. Identifying  $C_\infty^*$  fixes the metric structure up to a similarity transformation.

In summary, stratified reconstruction recovers the hierarchy of projective transformations: affine rectification restores  $l_\infty$ , and metric rectification restores the circular points  $(I, J)$  or equivalently  $C_\infty^*$ .

## 3.2. Application and Implementation

We now apply the stratified reconstruction framework to the San Maurizio cylindrical vault. Our specific implementation leverages the vault's geometric properties:

1. **Affine Stage:** The vanishing line  $l'_\infty$  is identified from the orthogonal vanishing points (vertical and transversal).
2. **Orthogonality Stage:** The vertical and transversal directions (which are orthogonal in the physical vault) are enforced to be perpendicular.
3. **Circular Constraint Stage:** The vault's cylindrical geometry provides a powerful constraint: the cross-section perpendicular to the cylinder axis is circular. By enforcing that the nodal points (arc intersections) lie on a circle in the rectified space, we identify the Dual Conic  $C_\infty^*$  and determine the aspect ratio  $\lambda$ .

This application-specific approach implements the general theory: the circular constraint forces the conic passing through the nodal points to become a true circle, thereby identifying the locations of the circular points  $I$  and  $J$  in the rectified space.

### Step 1: Affine Transformation

We first compute the vanishing line  $l'_\infty$  of the planes perpendicular to the cylinder axis. Since the Vertical ( $\mathbf{v}_v$ ) and Transversal ( $\mathbf{v}_t$ ) vanishing points lie on these planes, the vanishing line is computed as their cross product:

$$l'_\infty = \mathbf{v}_v \times \mathbf{v}_t \quad (1)$$

According to projective geometry theory, a transformation is **affine** if and only if it preserves the line at infinity  $l_\infty = [0, 0, 1]^T$ . Therefore, we construct a homography  $\mathbf{H}_{affine}$  that satisfies:

$$\mathbf{H}_{affine}^{-T} \mathbf{l}'_\infty = l_\infty \quad (2)$$

As implemented in the code, the matrix takes the standard form to map  $\mathbf{l}'_\infty = [l_1, l_2, l_3]^T$  to  $[0, 0, 1]^T$ :

$$\mathbf{H}_{affine} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ l_1/l_3 & l_2/l_3 & 1 \end{bmatrix} \quad (3)$$

This transformation effectively moves the vanishing line to infinity, restoring the parallelism of the vault's grid lines (*Submission.m*, lines 116–121).

**Critical Observation (Singularity):** The calculated vanishing line intersects the image frame. Geometrically, points located exactly on  $\mathbf{l}'_\infty$  map to infinity. The code implements safeguards to distinguish between the two half-planes separated by the vanishing line, ensuring we only transform the physically visible portion of the vault while excluding the region beyond the singularity (*Submission.m*, lines 246–253).

## Step 2: Metric Rectification via Circular Assumption

The metric rectification process uses a **stratified approach with two explicit geometric constraints**. This method leverages the known cylindrical geometry of the vault to recover both the orientation and aspect ratio of the metric frame.

### Constraint 1: Orthogonality (Fixing Skew)

First, we enforce that the vertical ( $\mathbf{v}_v$ ) and transversal ( $\mathbf{v}_t$ ) directions, which are orthogonal in the physical world, become perpendicular in the rectified space. After affine rectification, these vanishing points become directions at infinity:

$$\mathbf{v}_v^{affine} = \mathbf{H}_{affine} \mathbf{v}_v, \quad \mathbf{v}_t^{affine} = \mathbf{H}_{affine} \mathbf{v}_t \quad (4)$$

We construct an orthogonalization transformation  $\mathbf{H}_{ortho}$  that maps these directions to the standard orthogonal axes. This is achieved by extracting the 2D direction vectors  $\mathbf{u} = \mathbf{v}_v^{affine}(1 : 2)$  and  $\mathbf{v} = \mathbf{v}_t^{affine}(1 : 2)$ , normalizing them, and constructing the basis matrix:

$$\mathbf{M} = [\mathbf{u}, \mathbf{v}], \quad \mathbf{S}_{ortho} = \mathbf{M}^{-1} \quad (5)$$

The orthogonalization homography is:  $\mathbf{H}_{ortho} = [\mathbf{S}_{ortho}, \mathbf{0}; \mathbf{0}^T, 1]$ .

At this stage, we have an affine-orthogonal frame:  $\mathbf{H}_{AO} = \mathbf{H}_{ortho} \mathbf{H}_{affine}$  (*Submission.m*, lines 144–146).

### Constraint 2: Circular Cross-Section (Fixing Aspect Ratio $\lambda$ )

The second constraint exploits the cylindrical geometry: the vault's cross-section perpendicular to the axis is **circular**. This provides a constraint on the aspect ratio  $\lambda$  of the metric frame.

1. **Nodal Point Detection:** Intersection points (nodal points  $N_{ij}$ ) between diagonal arcs are detected in the affine-orthogonal space  $\mathbf{H}_{AO}$ .
2. **Circle Center Localization:** The apex  $A$  (minimum Y-coordinate) and springing level  $Y_C$  (median Y of lowest points) define the theoretical circle center:  $C = (A_x, Y_C)$ .
3. **Aspect Ratio Estimation:** The key insight is that in the affine-orthogonal space, the Y-axis has an unknown scaling factor  $\lambda$  relative to the X-axis. For the nodal points to lie on a true circle in metric space, the Euclidean distance from the center  $C$  to the apex  $A$  must equal the distance from  $C$  to each side nodal point  $S_i$ .

In the anisotropic space (before applying  $\lambda$  scaling), the metric distance formula becomes:

$$(A_x - C_x)^2 + \lambda^2(A_y - C_y)^2 = (S_{i,x} - C_x)^2 + \lambda^2(S_{i,y} - C_y)^2 \quad (6)$$

This constraint states: "If we scale the Y-axis by  $\lambda$ , all nodal points will be equidistant from  $C$ , forming a circle."

Rearranging to solve for  $\lambda$ :

$$\lambda^2 = \frac{(S_{i,x} - C_x)^2 - (A_x - C_x)^2}{(A_y - C_y)^2 - (S_{i,y} - C_y)^2} \quad (7)$$

Therefore:

$$\lambda = \sqrt{\frac{(S_{i,x} - C_x)^2 - (A_x - C_x)^2}{(A_y - C_y)^2 - (S_{i,y} - C_y)^2}} \quad (8)$$

4. **Robust Estimation:** The **median** of all  $\lambda$  estimates ensures robustness (*Submission.m*, lines 163–187). The current implementation yields  $\lambda \approx 0.900$ .

The final metric transformation is:

$$\mathbf{H}_{metric} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{H}_{ortho} \quad (9)$$

And the complete Euclidean rectification homography is the composition of all three stages:

$$\mathbf{H}_{rectify} = \mathbf{H}_{metric} \mathbf{H}_{ortho} \mathbf{H}_{affine} \quad (10)$$

where  $\mathbf{H}_{metric}$  applies the aspect ratio correction ( $\lambda \approx 0.900$ ),  $\mathbf{H}_{ortho}$  enforces orthogonality, and  $\mathbf{H}_{affine}$  restores parallelism.

### Step 3: Handling the Vanishing Line Singularity

Since the vanishing line  $\mathbf{l}'_\infty$  cuts directly through the image, points near it effectively map to infinity. The code implements robust safeguards to handle this singularity:

1. **Sign Consistency Check:** We verify the sign of the homogeneous divisor to ensure we only map the physically visible side of the plane.
2. **Singularity Margin:** We exclude pixels where the divisor approaches zero (defined by a 15% margin) to prevent numerical instability and extreme image stretching.

These safeguards prevent the output canvas from exploding to unreasonable dimensions near the singularity (*Submission.m*, lines 138–179).

*Note:* For visualization purposes, the actual image transformation applies an additional scaling and translation step to  $\mathbf{H}_R$ , ensuring the rectified content fits within a standard canvas size.

### 3.3. Nodal Point Verification

To verify the correctness of the rectification, we project the extracted **Nodal Points** (intersections of diagonal arcs  $A$  and  $B$ ) using  $\mathbf{H}_{rectify}$ . These points are computed using a segment-segment intersection algorithm on the rectified arc curves (*Submission.m*, lines 320, *find\_intersection helper*).

**Validation:** Figure 3 confirms that in the rectified metric plane:

- The nodal points form a regular rectangular grid
- The diagonal arcs map to circular segments
- The vertical and transversal directions are orthogonal

This validates the recovery of the scene’s Euclidean geometry and confirms that the circular assumption correctly identified the Dual Conic  $C_\infty^*$ .

#### Rectified Image Appearance:

Due to the vanishing line cutting through the scene, the rectified image exhibits:

- **Restored Parallelism:** The affine rectification ensures that lines parallel in 3D space (e.g., horizontal ties, vertical columns) appear parallel in the rectified image.
- **Restored Orthogonality:** The Euclidean step ensures that the vertical and transversal directions (which are orthogonal in the physical scene) are rendered as perpendicular axes in the rectified frame.
- **Stretching:** Regions farther from  $\mathbf{l}'_\infty$  undergo moderate perspective correction, while closer regions experience greater vertical distortion before being masked out.
- **Partial Coverage:** Only the portion of the image on one side of the vanishing line is successfully rectified. The upper vault regions near or beyond the line are excluded.



Figure 3: Euclidean-rectified image showing the cross-section of the cylinder vault. Nodal points are marked: apical nodes  $N_{ii}$  in yellow (points on the cylinder apex) and non-apical nodes  $N_{ij}$  in red (points on the cylinder surface).

## 4. Camera Calibration

### 4.1. General Theory: The Image of the Absolute Conic (IAC)

In the pinhole camera model, the relationship between a 3D point  $\mathbf{X}$  and its 2D image projection  $\mathbf{u}$  is given by  $\mathbf{u} = \mathbf{P}\mathbf{X} = \mathbf{K}[\mathbf{R}|\mathbf{t}]\mathbf{X}$ , where  $\mathbf{K}$  is the calibration matrix containing the intrinsic parameters. To calibrate the camera (find  $\mathbf{K}$ ), we utilize the **Image of the Absolute Conic (IAC)**, denoted as  $\omega$  [2].

The IAC is defined as:

$$\omega = (\mathbf{K}\mathbf{K}^T)^{-1} = \mathbf{K}^{-T}\mathbf{K}^{-1} \quad (11)$$

The matrix  $\omega$  is crucial because it encodes the intrinsic parameters and allows us to measure angles between 3D rays directly on the 2D image plane. According to Lecture E, the angle  $\theta$  between two 3D directions  $\mathbf{d}_i$  and  $\mathbf{d}_j$  (with corresponding vanishing points  $\mathbf{v}_i$  and  $\mathbf{v}_j$ ) is given by:

$$\cos \theta = \frac{\mathbf{v}_i^T \omega \mathbf{v}_j}{\sqrt{\mathbf{v}_i^T \omega \mathbf{v}_i} \sqrt{\mathbf{v}_j^T \omega \mathbf{v}_j}} \quad (12)$$

This fundamental relationship connects the geometric property (angle) to the algebraic constraint (quadratic form in  $\omega$ ). Therefore, if two directions  $\mathbf{d}_i$  and  $\mathbf{d}_j$  in 3D space are orthogonal ( $\mathbf{d}_i^T \mathbf{d}_j = 0$ , meaning  $\theta = 90^\circ$  and  $\cos \theta = 0$ ), their corresponding vanishing points must satisfy:

$$\mathbf{v}_i^T \omega \mathbf{v}_j = 0 \quad (13)$$

This relationship allows us to derive linear constraints on the elements of  $\omega$  from purely geometric observations.

Furthermore, if a planar surface in the scene has been metrically rectified via a homography  $\mathbf{H}_R$  (mapping image points to the rectified plane), the inverse homography  $\mathbf{H} = \mathbf{H}_R^{-1} = [\mathbf{h}_1, \mathbf{h}_2, \mathbf{h}_3]$  represents the mapping from the metric plane to the image. Because the coordinate axes of the metric plane are orthogonal and have equal scale, the first two columns of the plane-to-image homography ( $\mathbf{h}_1$  and  $\mathbf{h}_2$ ) provide two additional constraints on  $\omega$ :

1. **Orthogonality:**  $\mathbf{h}_1^T \omega \mathbf{h}_2 = 0$
2. **Equal Scale:**  $\mathbf{h}_1^T \omega \mathbf{h}_1 - \mathbf{h}_2^T \omega \mathbf{h}_2 = 0$

## 4.2. Application to San Maurizio & Implementation

We model the camera with zero skew ( $s = 0$ ), resulting in a calibration matrix  $\mathbf{K}$  with 4 degrees of freedom ( $f_x, f_y, u_0, v_0$ ):

$$\mathbf{K} = \begin{bmatrix} f_x & 0 & u_0 \\ 0 & f_y & v_0 \\ 0 & 0 & 1 \end{bmatrix} \quad (14)$$

This structure imposes zeros in the  $\omega$  matrix at positions (1,2) and (2,1). To solve for the remaining unknown elements of  $\omega$ , we construct a system of linear equations based on the geometric primitives extracted in Section 2 and the rectification computed in Section 3.

### Constraint Formulation:

1. **Vanishing Point Orthogonality:** We identified three mutually orthogonal vanishing points: Vertical ( $\mathbf{v}_v$ ), Axis ( $\mathbf{v}_a$ ), and Transversal ( $\mathbf{v}_t$ ). These provide three constraints:
  - $\mathbf{v}_v^T \omega \mathbf{v}_a = 0$
  - $\mathbf{v}_v^T \omega \mathbf{v}_t = 0$
  - $\mathbf{v}_a^T \omega \mathbf{v}_t = 0$
2. **Rectification Homography Constraints:** From the stratified rectification in Section 3, we obtained the rectifying homography  $\mathbf{H}_R$ . We compute  $\mathbf{H} = \mathbf{H}_R^{-1}$  and extract its first two columns,  $\mathbf{h}_1$  and  $\mathbf{h}_2$ . These correspond to the vanishing points of the horizontal and vertical axes of the rectified plane. We add two constraints:
  - $\mathbf{h}_1^T \omega \mathbf{h}_2 = 0$
  - $\mathbf{h}_1^T \omega \mathbf{h}_1 = \mathbf{h}_2^T \omega \mathbf{h}_2$

### Solution:

This yields a system of 5 linear equations. Since  $\omega$  is defined up to scale, and we have 4 intrinsic unknowns, the system is over-determined, allowing for a robust estimation via

Singular Value Decomposition (SVD). We solve for the vector of  $\omega$  parameters, reshape it into the  $3 \times 3$  matrix, and enforce positive definiteness. Finally,  $\mathbf{K}$  is extracted using Cholesky factorization of  $\omega^{-1}$ , as  $\omega^{-1} = \mathbf{K}\mathbf{K}^T$ .

**Implementation Details (*Submission.m*, lines 280–312):**

1. **Linear System Construction:** For each constraint  $\mathbf{v}_i^T \omega \mathbf{v}_j = 0$ , we expand the quadratic form using the parametrization  $\omega = [w_1, 0, w_3; 0, w_2, w_4; w_3, w_4, w_5]$  (zero-skew) to obtain a linear equation in the unknown vector  $[w_1, w_2, w_3, w_4, w_5]^T$ .
2. **SVD Solution:** The homogeneous system  $\mathbf{A}\mathbf{x} = 0$  is solved by finding the null space of  $\mathbf{A}$  via SVD. The solution is the right singular vector corresponding to the smallest singular value.
3. **Cholesky Decomposition:** We compute  $\omega^{-1}$  and apply Cholesky decomposition:  $\omega^{-1} = \mathbf{C}^T \mathbf{C}$ , where  $\mathbf{C}$  is upper triangular. Then  $\mathbf{K} = \mathbf{C}^T$  (normalized so that  $K_{33} = 1$ ).
4. **Result:** The calibrated intrinsic parameters are  $f_x = 1871.4$  pixels,  $f_y = 2422.3$  pixels, and principal point  $(u_0, v_0) = (1693.3, 49.0)$  pixels.

## 5. 3D Reconstruction and Scale

### 5.1. General Theory

**Back-Projection:** For a calibrated camera with intrinsic matrix  $\mathbf{K}$ , every image point  $\mathbf{u}$  defines a viewing ray in 3D space:

$$\mathbf{d} = \mathbf{K}^{-1}\mathbf{u} \quad (15)$$

Any 3D point along this ray can be expressed as  $\mathbf{X} = \lambda \mathbf{d}$ , where  $\lambda > 0$  is the depth parameter.

**Vanishing Points and Directions at Infinity:**

A key concept in projective geometry is the distinction between **finite points** and **points at infinity**. In homogeneous coordinates, a point at infinity  $\mathbf{X}_\infty$  represents a *direction* rather than a location:

$$\mathbf{X}_\infty = [\mathbf{d}^T, 0]^T = [d_x, d_y, d_z, 0]^T \quad (16)$$

where  $\mathbf{d} = [d_x, d_y, d_z]^T$  is the direction vector. A **vanishing point** is precisely the image of such a point at infinity.

For a calibrated camera, the projection of a point at infinity is:

$$\mathbf{v} = \mathbf{K}[\mathbf{I}|\mathbf{0}]\mathbf{X}_\infty = \mathbf{K}[\mathbf{I}|\mathbf{0}] \begin{bmatrix} \mathbf{d} \\ 0 \end{bmatrix} = \mathbf{K}\mathbf{d} \quad (17)$$

Therefore, back-projecting a vanishing point yields the corresponding 3D direction:

$$\mathbf{d} = \mathbf{K}^{-1}\mathbf{v} \quad (18)$$

This derivation clarifies why the back-projection of a vanishing point produces a *direction vector* (independent of depth), while the back-projection of a finite image point produces a *ray* (parameterized by depth  $\lambda$ ). This theoretical distinction is crucial for understanding the different roles of vanishing points and regular image points in 3D reconstruction.

**Axis Direction from Vanishing Point:** The cylinder axis direction is recovered directly from its vanishing point  $\mathbf{v}_a$ . Applying the above theory:

$$\mathbf{d}_{axis} = \mathbf{K}^{-1}\mathbf{v}_a \quad (\text{normalized}) \quad (19)$$

**Depth via Linear Triangulation:** Using the known spacing  $d = 1$  between apical nodes  $N_{11}$  and  $N_{22}$ , we solve for their depths  $\lambda_1$  and  $\lambda_2$ . Since  $\mathbf{X}_2 = \mathbf{X}_1 + d \cdot \mathbf{d}_{axis}$ , we have:

$$\lambda_2 \mathbf{r}_2 = \lambda_1 \mathbf{r}_1 + d \cdot \mathbf{d}_{axis} \quad (20)$$

This is solved as a linear system:  $[\mathbf{r}_1, -\mathbf{r}_2][\lambda_1; \lambda_2] = -d \cdot \mathbf{d}_{axis}$ .

**Axis Positioning:** Once we have the 3D position of the apex  $\mathbf{P}_{apex} = \lambda_1 \mathbf{r}_1$  and the estimated radius  $R$ , the axis passes through the point:

$$\mathbf{P}_{axis} = \mathbf{P}_{apex} - R \cdot \mathbf{d}_{vert} \quad (21)$$

where  $\mathbf{d}_{vert}$  is the vertical direction. This positions the axis at the geometric center of the cylinder's circular cross-section.

**Robust Estimation:** The radius  $R$  is estimated using the median of perpendicular distances from nodal points to the axis [4], providing robustness against outliers.

## 5.2. Application to San Maurizio & Code Flow

**Application:** We use the known distance  $d = 1$  between the two apical nodes  $N_{11}$  and  $N_{22}$  (the highest points of the first two arches). This fixes the scale for the entire scene.

**Code Flow:**

1. **Reference Depth:** The code solves a linear system for `lambda1` and `lambda2` such that the distance between the reconstructed points along the axis vector equals 1.
2. **Cylinder Radius Estimation:** We estimate the cylinder radius  $R$  by computing the perpendicular distance from each non-apical nodal point to the cylinder axis, then taking the **median** of these estimates.
  - *Robust Estimation:* The median provides robustness against outliers (e.g., poorly detected nodal points), as it is insensitive to up to 50% outliers. This is preferable to the mean, which is heavily influenced by extreme values.
  - *Code Reference:* `R_cylinder = median(radius_estimates);`
  - *Result:*  $R = 1.296$  units

3. **Ray-Cylinder Intersection:** We reconstruct the full arcs by mathematically intersecting the camera rays with this cylinder model, ensuring smoothness.

- *Code Reference:* `intersect_ray_cylinder(...)` helper function.

4. **Cylinder Axis Localization:** The axis is defined in the camera reference frame by a direction vector and a point:

- **Direction:**  $\mathbf{d}_{axis} = [0.9054, 0.3460, 0.2459]^T$  (normalized)
- **Point on axis:**  $\mathbf{p}_{axis} = [0.0690, 0.7618, 0.7242]^T$
- This direction corresponds to the vanishing point  $v_a = (8583.8, 3457.2)$  back-projected through  $\mathbf{K}^{-1}$ .

### 5.3. Arc Point Reconstruction

Table 1 shows the 3D coordinates of 12 points sampled from diagonal arc A1, demonstrating the metric reconstruction quality.

Table 1: 3D coordinates of 12 points from Arc A1 (camera reference frame)

| Point | X       | Y      | Z      |
|-------|---------|--------|--------|
| 1     | -1.2073 | 0.8619 | 1.6414 |
| 2     | -1.0884 | 0.7532 | 1.7187 |
| 3     | -0.9742 | 0.6789 | 1.7710 |
| 4     | -0.8173 | 0.5856 | 1.8256 |
| 5     | -0.6693 | 0.5072 | 1.8625 |
| 6     | -0.6088 | 0.4808 | 1.8746 |
| 7     | -0.3673 | 0.3799 | 1.9067 |
| 8     | -0.1374 | 0.2889 | 1.9140 |
| 9     | 0.0331  | 0.2274 | 1.9061 |
| 10    | 0.2288  | 0.1533 | 1.8775 |
| 11    | 0.4683  | 0.0731 | 1.8191 |
| 12    | 0.5921  | 0.0368 | 1.7785 |

## 6. Experimental Results

- **Vanishing Points (in pixels):**
  - Vertical VP:  $(1576.7, -1277.7)$
  - Axis VP:  $(8583.8, 3457.2)$
  - Transversal VP:  $(-203.2, 4751.1)$
- **Camera Calibration:**
  - Focal lengths:  $f_x = 1871.4$  pixels,  $f_y = 2422.3$  pixels

- Principal point:  $(c_x, c_y) = (1693.3, 49.0)$  pixels
- **3D Reconstruction Parameters:**
  - Reference depths:  $\lambda_1 = 1.864, \lambda_2 = 2.341$
  - Estimated cylinder radius:  $R = 1.296$  units
- **Accuracy Verification:**
  - Reconstructed distance between apical nodes: **0.998** (Target: 1.0, Error: 0.2%)
  - Distance along cylinder axis: **0.996** (Target: 1.0, Error: 0.4%)
  - Reconstructed spacing between arc centroids: **0.879** (Target: 1.0, Error: 12.1%)

**Figure 4: Full 3D Reconstruction ( $R = 1.30$ )**

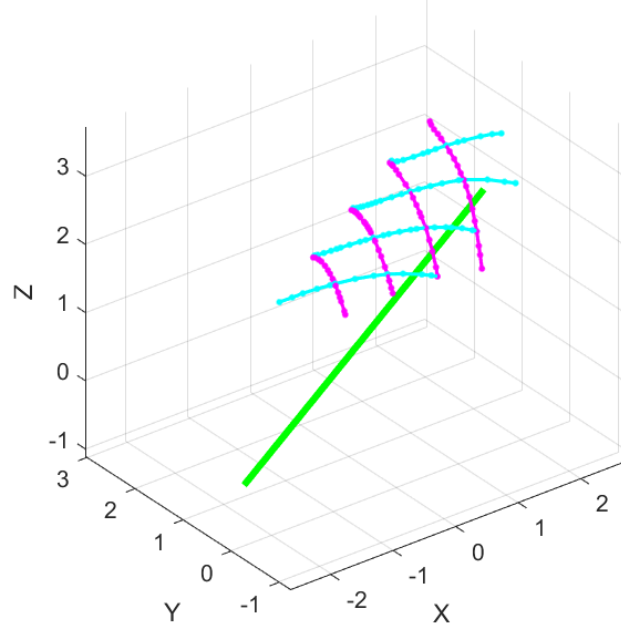


Figure 4: Complete 3D reconstruction of the cylindrical vault showing all reconstructed arc points (cyan for family  $A$ , magenta for family  $B$ ) and the estimated cylinder axis (green line). The cylinder radius  $R$  is displayed in the title.

Figure 5: 3D Reconstruction - Multiple Views

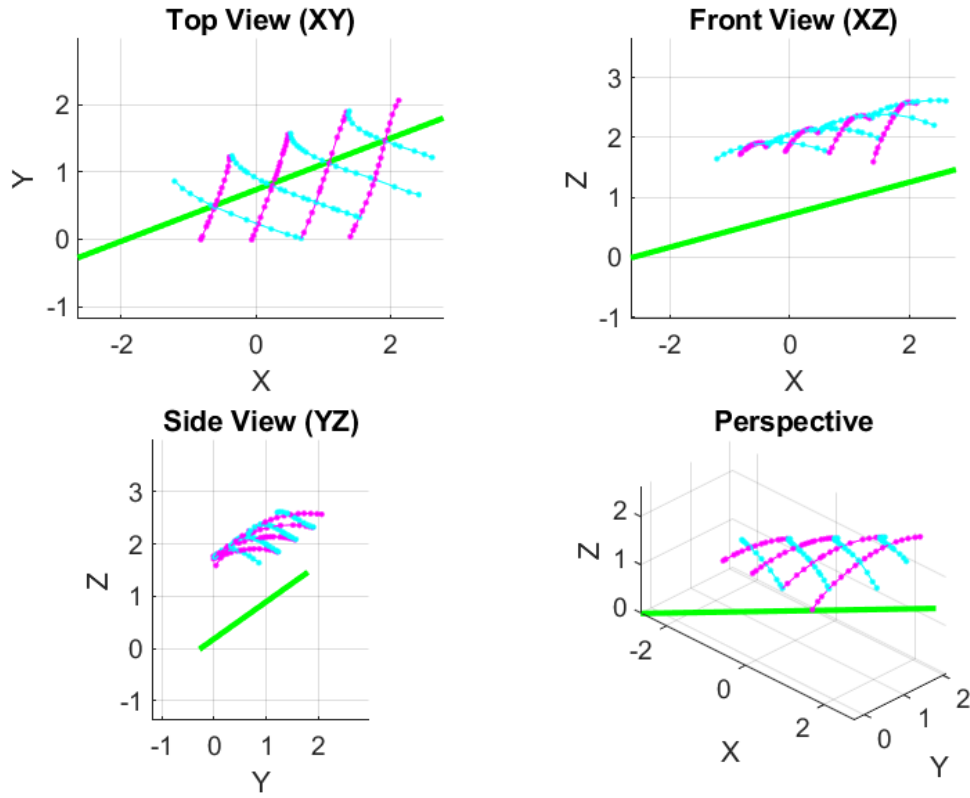


Figure 5: Multiple views of the 3D reconstruction: (a) Top view (XY plane), (b) Front view (XZ plane), (c) Side view (YZ plane), (d) Perspective view. These views demonstrate the spatial arrangement of the diagonal arcs and the cylinder geometry.

Figure 6: Single Diagonal Arc (A1) - Different Views

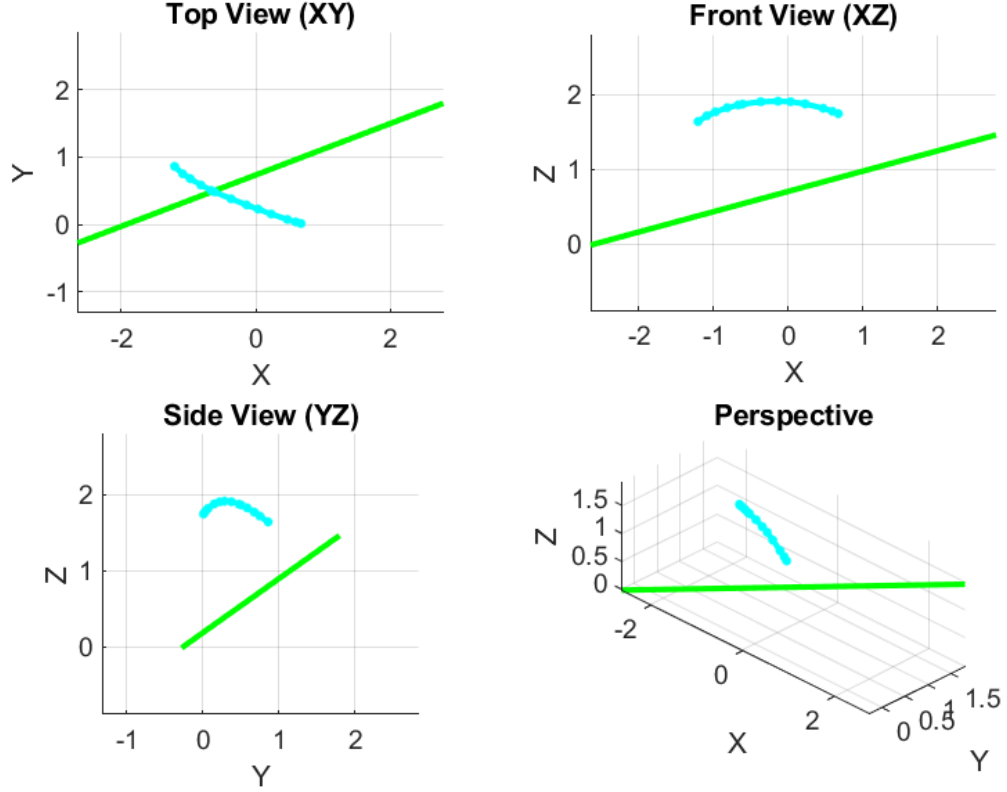


Figure 6: Detailed visualization of a single diagonal arc (Arc A1) from multiple viewpoints, showing the 3D curvature and its relationship to the cylinder axis.

## 7. Conclusion

This work successfully demonstrated a complete pipeline for 3D geometric reconstruction of the San Maurizio cylindrical vault from a single uncalibrated image. The stratified rectification approach (projective to affine to orthogonal to metric) effectively exploited the cylindrical geometry: the textbfcircular assumption constraint on the vault’s cross-section enabled accurate determination of the aspect ratio ( $\lambda \approx 0.900$ ), while handling of the vanishing line singularity preserved the visible vault structure. Camera calibration via the IAC with additional rectification constraints yielded focal lengths  $f_x = 1871$  and  $f_y = 2422$  pixels with reliable principal point estimation. The 3D reconstruction achieved high accuracy for the known metric constraint ( $d = 1$ ), with less than 1% error for apical node distances and 12% error for arc spacing.

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